

Never Start From Scratch: Expediting On-Device LLM Personalization via Explainable Model Selection

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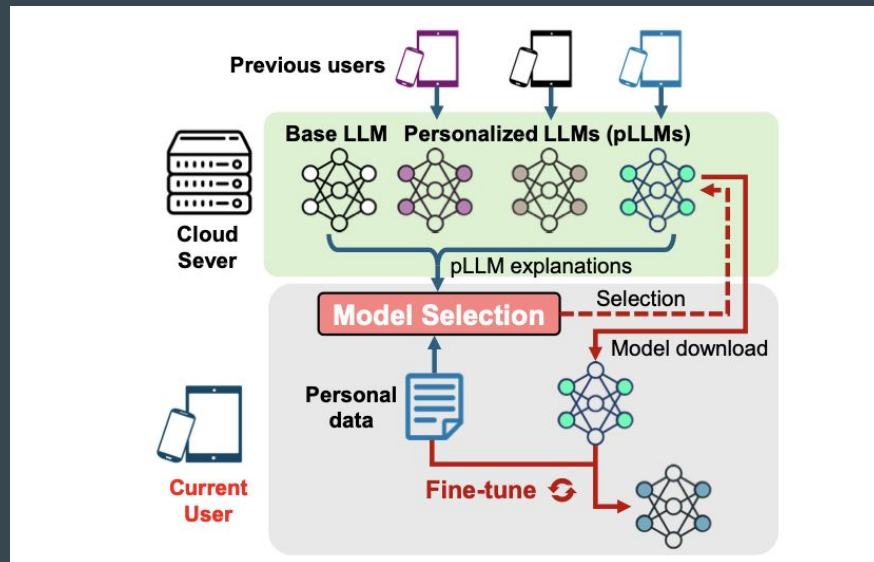
Haoming Wang

The Problem

- On-Device LLMs are compute constrained
- Personalization requires fine tuning
- Full retraining is slow
- Sending data to cloud breaks privacy

Reuse Similar Personalized Models

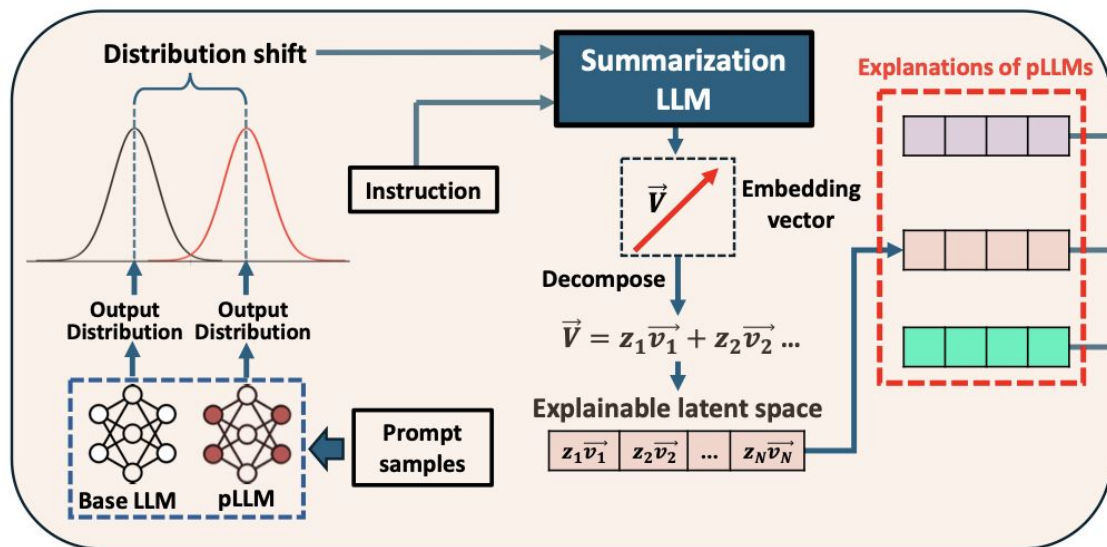
- Don't fine-tune from scratch on every device
- Reuse previously personalized models store in the cloud
- Find the model whose behavior best matches user
- Perform small local refinement on device



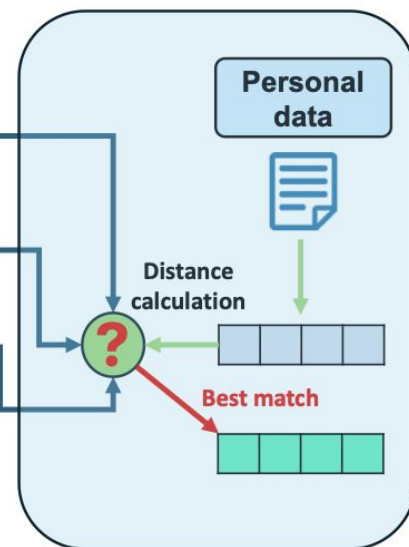
Explainable Model Selection

- Compare how base and personalized models behave
- Describe differences using interpretable attributes
- Match attributes to cached models
- Download best match

System Design



Model Explanation on the Server



On-Device Model Selection

Results

- Up to 90% reduction in training steps
- Significant reduction in required training data
- Higher similarity = greater efficiency gains

